

Job Board with Matching Algorithm

Objectives

- Develop an advanced job board matching candidates to relevant positions using skill-based scoring.
- Implement resume parsing to extract skills, experience, and education automatically.
- Provide application tracking for candidates and a hiring pipeline dashboard for employers.
- Enable employers to post jobs with structured requirements and view ranked applicant lists.

Deliverables

- A fully functional job board platform for both candidates and employers.
- Resume upload and parser extracting structured data from PDF and DOCX files.
- Skill-based matching engine scoring candidate-job fit as a percentage.
- Candidate dashboard tracking application statuses (applied, reviewed, interviewed, offered).
- Employer dashboard with job posting management, applicant ranking, and interview scheduling.

Requirements

- Frontend: React.js with a clean, accessible interface for both user roles.
- Backend: Node.js/Express REST API with role-based access control (candidate vs employer).
- Resume Parsing: Python service using spaCy NER to extract skills, education, and experience.
- Matching: Cosine similarity scoring between job requirement vectors and candidate skill vectors.
- Database: PostgreSQL for relational job, user, and application data.

Advanced Features

- Resume Parsing: NLP-powered extraction supporting PDF, DOCX, and plain text formats.
 - Skill-Based Matching: Weighted scoring algorithm accounting for must-have vs nice-to-have skills.
 - Application Tracking: Kanban-style pipeline view for employers to move candidates through stages.
 - Email Notifications: Automated status updates sent to candidates at each pipeline stage.
 - Saved Searches: Candidates save job filters and receive alerts when matching roles are posted.
 - Company Profiles: Employer branding pages with culture, benefits, and team photos.
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- Interview Scheduler: Integrated calendar availability picker syncing with Google Calendar.
 - Salary Insights: Aggregated market salary data displayed per job category and location.
 - Analytics Dashboard: Employer metrics on job views, application rates, and time-to-hire.
 - Mobile Responsiveness: Fully responsive design for job searching and applying on mobile devices.
 - Payment Gateway: Charge employers for premium job postings and featured listings via Stripe.
 - Candidate Subscription: Offer candidates a paid plan for advanced filters, resume boosts, and priority applications.
 - Invoice on Hire: Optionally generate a placement invoice for recruitment agencies upon candidate hire.